

Ahmed Farag

Data Scientist | Machine Learning Engineer | AI Engineer

Alexandria, Egypt | af0112651@gmail.com | +201120577817 | +201126516608

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PROFESSIONAL SUMMARY

Recent Computer and Data Science graduate with hands-on experience in Machine Learning, Artificial Intelligence, Data Analysis, and Backend Development. Skilled in Python, SQL, FastAPI, TensorFlow, PyTorch, and Scikit-learn, with experience building recommendation systems, predictive models, and AI-powered applications.

PROJECTS

1. Property Hub AI (Graduation Project)

AI-Powered Real Estate Platform

- Developed a full-stack AI-powered real estate platform using React, FastAPI, MySQL design 5 Table, and Redis.
- Built a hybrid recommendation system using Content-Based and Learning-to-Rank techniques use 200 users.
- Trained a property price prediction model on 26,000+ property listings, achieving an R^2 score of 86%.
- Designed and integrated RESTful APIs for authentication, property search, 2 model recommendations, analytics, and price prediction.👉

2. Traffic Analytics

Big Data Analytics Platform

- Built a traffic analytics platform using Apache Spark, Apache Kafka, and Streamlit.
- Processed and analyzed 10,000 simulated traffic records to extract traffic patterns and congestion insights.
- Designed interactive dashboards for visualizing traffic flow and simulation results.
- Implemented a scalable big data pipeline for efficient traffic data processing.👉

3. Customer Churn Prediction

Machine Learning Prediction System

- Implemented a machine learning solution to predict customer churn using Python and Scikit-learn.
- Performed data preprocessing, feature engineering, and class balancing (SMOTE) on 7,043 customer records.
- Trained and evaluated 5 classification models, achieving 80.3% prediction accuracy with the best-performing model.
- Evaluated model performance using Accuracy, Precision, Recall, and F1-score to support customer retention analysis.👉

EDUCATION

Alexandria University

Bachelor of Computer and Data Science

Cumulative GPA: 3.43/4.0 Sept, 2022- Jul 202

TECHNICAL SKILLS

Programming: Python, SQL, Java, C

Machine Learning & AI: Scikit-learn, TensorFlow, PyTorch, Keras, Deep Learning, Recommendation Systems, Feature Engineering, Model Evaluation, Prediction System.

Data Science: Pandas, NumPy, Data Analysis, Data Cleaning, Data Preprocessing, EDA, Statistics, Data Visualization

Databases: MySQL, MongoDB, PostgreSQL, SQLAlchemy

Backend: FastAPI, REST APIs, JWT Authentication, Redis

Big Data: Apache Hadoop, HDFS, Apache Kafka, Apache Spark

Tools: Git, GitHub, VS Code, Jupyter Notebook, Postman, Docker

PROFESSIONAL DEVELOPMENT

- Machine Learning Training — National Telecommunication Institute (NTI)
- AI & Machine Learning Summer Training — MAIM Company
- Fundamentals of Deep Learning — NVIDIA
- Introduction to Data Science — CISCO
- Computer Networks Training — Abu Qir Electricity Distribution Company
- IoT Summer Training — Faculty of Computers and Data Science